



Jean-Baptiste Besnard

LEAD SOFTWARE ENGINEER

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Experience

Lead Software Engineer

Meudon, France

DATA DIRECT NETWORKS (DDN)

Jan. 2025 - present

- Lead efforts on a high-performance client for Infinia leveraging its SDK
- Designed the high-frequency C++ monitoring infrastructure inside Infinia code, bridging it with standard collectors

Manager and Technical Leader

Bruyère-le-Château, France

PARATOOLS SAS

Sept. 2016 - Dec. 2024

- Lead a team of consultants to timely meet their objectives
- Identify and conduct innovative R&D studies and developments
- Respond to public calls
- Interact with clients and write proposals
- Technical assistance to teams

Expert in High-Performance Computing (HPC)

Bruyère-le-Château, France

PARATOOLS SAS

Aug. 2014 - Sept. 2016

- Development of a thread-based MPI implementation for supercomputers
- Work on high-speed networks for High-Performance Computing
- Extension of GCC to add a new TLS level

Research Engineer

Versailles, France

EXASCALE COMPUTING RESEARCH

Oct. 2013 - Aug. 2014

- Develop innovative web-based interfaces for performance analysis
- Use of D3.js in frontend and Node.js in backend

Ph.D. Student

Bruyères-le-Château, France

CEA, DAM, DIF

Oct. 2010 - Jul. 2013

- Development of an original solution for massive performance trace data-management
- Using in-situ processing techniques for in-flight data-processing
- Validation at super-computer scale (Tera 100 at the time)

Skills

Programming	C, Rust, C++, Python, JavaScript (Node.js), Go, Ba(sh)
Networking	TCP, MPI, Infiniband, Libfabric, gRPC
Scalable Performance Monitoring	Prometheus, Grafana, Open-Telemetry, code instrumentation, custom data-reduction
Parallelism	MPI, OpenMP, CUDA, Pthread, Profiling
Administration	Linux, Nginx, Reverse Proxy, Docker, IPTables, Networking
Project Management	Write calls, software project management, client facing, budget management
Research	Paper redaction, Respond to calls, Standardization committees (MPI)
Languages	French (native), English

Projects

DDN Infinia

C, C++, Rust, I/O, Performance

OBSERVABILITY AND RUNTIME PROGRAMMING

<https://www.ddn.com/products/infinia/>

- Observability and Debuggability of Infinia (Prometheus, Open Telemetry (C++))
- Implemented a high-performance log analyzer (Rust)
- Work on high-performance client service for I/O (C++/C)

ADMIRE Euro-HPC Project

PARALLEL PERFORMANCE MONITORING AT SCALE

- Implementation of a distributed always on monitoring infrastructure at supercomputer scale for network and I/O (in Rust)
- Interfacing with existing tools (Grafana and Prometheus)
- Write scientific papers and do reporting including financial

C, MPI, Rust, I/O, Performance

admire-eurohpc.eu

Muti-Processor Computing

A THREAD-BASED UNIFIED IMPLEMENTATION OF MPI AND OPENMP

- Low level development for high-speed networks on supercomputers (libfabric, Infiniband)
- Implementation of RDMA and SHM
- Work in GCC compiler for automatic privatization
- Work on scalability of collective communications

C, Fortran, MPI, OpenMP

hpcframework.com/mpc/

PCOCC

PRIVATE CLOUD ON A COMPUTE CLUSTER

- Work in a virtual machine orchestrator dedicated to HPC payloads
- Development in GO of an agent for VM
- Extension of the orchestrator to support containers (OCI images)
- Integration of Docker inside a secure environment (inside a VM managed by the orchestrator)

Python, OCI, Docker, Go, gRPC

pcocc.readthedocs.io

Muti-Application Online Profiling

AN IN-SITU PROFILING TOOL FOR MPI APPLICATIONS

- Runtime coupling (instead of I/Os) to process data in-flight through network coupling
- Instrumentation of POSIX I/O and MPI calls
- Processing of data at scale (100s of TB)
- Visualization with an interactive HTML interface (Node.js and HTML)

C, MPI, OpenMP, Node.js, HTML

hpcframework.com/malp/

Teaching

Teaching Module Lead

UNIVERSITÉ DE VERSAILLES SAINT-QUENTIN (UVSQ)

- Courses on Operating Systems internal architecture for HPC Students
- Course on introduction to Software engineering (engineering school students)

Versailles, France

Jan. 2019 - Dec. 2024

Teaching Assitant

INSTITUT DES SCIENCES ET TECHNIQUES DES YVELINES (ISTY)

- Processor architecture for second year engineering students

Velizy, France

Oct. 2013 - Jul. 2014

Teaching Assistant

UNIVERSITÉ DE VERSAILLES SAINT-QUENTIN (UVSQ)

- Advanced compilation techniques (M2)

Versailles, France

Oct. 2012 - Jul. 2013

Teaching Assistant

UNIVERSITÉ DE VERSAILLES SAINT-QUENTIN (UVSQ)

- C programming and UNIX environments (M1)
- Parallel optimization and profiling using MPI and OpenMP (M1)

Versailles, France

Oct. 2011 - Jul. 2012

Education

Ph.D. in Computer Science

CEA, DAM, DIF

- Profiling and debugging of massively parallel applications.

Bruyères-le-Châtel, France

Oct. 2010 - Sept. 2013

Engineering Diploma

INSTITUT SUPÉRIEUR DE L'ÉLECTRONIQUE ET DU NUMÉRIQUE (ISEN)

- Specialization in Telecommunication and Networks.

Brest, France

Sept. 2007 - Jun. 2010

Preparatory Mathematics Classes

INSTITUT SUPÉRIEUR DE L'ÉLECTRONIQUE ET DU NUMÉRIQUE (ISEN)

- Specialization in Telecommunication and Networks.

Brest, France

Sept. 2005 - Jun. 2007

- J. Adam, J.-B. Besnard, A. Roussel, J. Jaeger, P. Carribault, and M. Perache. To Share or Not to Share: a case for MPI in Shared-Memory. In *Proceedings of the EuroMPI24/Australia 2024 conference*, 2024. In press.
- Jean-Baptiste Besnard, Martin Schreiber, and Allen D. Malony. Towards a scale invariant syntax for dynamic job-level workflows. In *DYNRES HPC Workshop, EuroPAR24*, 2024. In press.
- Jean-Baptiste Besnard, Ahmad Tarraf, Alberto Cascajo, and Sameer Shende. Introducing the metric proxy for holistic i/o measurements (in press). In *HPC-IODC'24: Proceedings of the 10th HPC I/O in the Data Center Workshop*, 2024.
- Ahmad Tarraf, Martin Schreiber, Alberto Cascajo, Jean-Baptiste Besnard, Marc-André Vef, Dominik Huber, Sonja Happ, André Brinkmann, David E Singh, Hans-Christian Hoppe, et al. Malleability in modern hpc systems: Current experiences, challenges, and future opportunities. *IEEE Transactions on Parallel and Distributed Systems*, 2024. doi: 10.1109/TPDS.2024.3406764.
- Julien Adam, Jean-Baptiste Besnard, Paul Canat, Hugo Taboada, Adrien Roussel, Marc Pérache, Julien Jaeger, and Sameer Shende. Generating and scaling a multi-language test-suite for MPI. In *Proceedings of the 30th European MPI Users' Group Meeting, EuroMPI 2023*, Bristol, United Kingdom, September 11-13, 2023, pages 11:1–11:10. ACM, 2023. doi: 10.1145/3615318.3615329.
- Jean-Baptiste Besnard, Ahmad Tarraf, Clément Barthélemy, Alberto Cascajo, Emmanuel Jeannot, Sameer Shende, and Felix Wolf. Towards smarter schedulers: Molding jobs into the right shape via monitoring and modeling. In Amanda Bienz, Michèle Weiland, Marc Baboulin, and Carola Kruse, editors, *High Performance Computing - ISC High Performance 2023 International Workshops*, Hamburg, Germany, May 21-25, 2023, Revised Selected Papers, volume 13999 of *Lecture Notes in Computer Science*, pages 68–81. Springer, 2023. doi: 10.1007/978-3-031-40843-4_6.
- Jesús Carretero, Javier García Blas, André Brinkmann, Marc-André Vef, Jean-Baptiste Besnard, Massimo Torquati, Yi Ju, and Raffaele Montella. Adaptive HPC input/output systems. In Demetris Zeinalipour, Dora Blanco Heras, George Pallis, Herodotos Herodotou, Demetris Trihinas, Daniel Balouek, Patrick Diehl, Terry Cojean, Karl Furlinger, Maja Hanne Kirkeby, Matteo Nardelli, and Pierangelo di Sanzo, editors, *Euro-Par 2023: Parallel Processing Workshops - Euro-Par 2023 International Workshops*, Limassol, Cyprus, August 28 - September 1, 2023, Revised Selected Papers, Part II, volume 14352 of *Lecture Notes in Computer Science*, pages 199–202. Springer, 2023. doi: 10.1007/978-3-031-48803-0_16.
- Jesús Carretero, Javier García-Blas, Marco Aldinucci, Jean-Baptiste Besnard, Jean-Thomas Acquaviva, André Brinkmann, Marc-André Vef, Emmanuel Jeannot, Alberto Miranda, Ramon Nou, Morris Riedel, Massimo Torquati, and Felix Wolf. Adaptive multi-tier intelligent data manager for exascale. In Andrea Bartolini, Kristian F. D. Rietveld, Catherine D. Schuman, and Jose Moreira, editors, *Proceedings of the 20th ACM International Conference on Computing Frontiers, CF 2023*, Bologna, Italy, May 9-11, 2023, pages 285–290. ACM, 2023. doi: 10.1145/3587135.3592174.
- Jeff R. Hammond, Lisandro Dalcín, Erik Schnetter, Marc Pérache, Jean-Baptiste Besnard, Jed Brown, Gonzalo Brito Gadeschi, Simon Byrne, Joseph Schuchart, and Hui Zhou. MPI application binary interface standardization. In *Proceedings of the 30th European MPI Users' Group Meeting, EuroMPI 2023*, Bristol, United Kingdom, September 11-13, 2023, pages 1:1–1:12. ACM, 2023. doi: 10.1145/3615318.3615319.
- Hugo Taboada, Romain Pereira, Julien Jaeger, and Jean-Baptiste Besnard. Towards achieving transparent malleability thanks to MPI process virtualization. In Amanda Bienz, Michèle Weiland, Marc Baboulin, and Carola Kruse, editors, *High Performance Computing - ISC High Performance 2023 International Workshops*, Hamburg, Germany, May 21-25, 2023, Revised Selected Papers, volume 13999 of *Lecture Notes in Computer Science*, pages 28–41. Springer, 2023. doi: 10.1007/978-3-031-40843-4_3.
- Jean-Baptiste Besnard, Sameer Shende, Allen Malony, Julien Jaeger, and Marc Perache. Enabling global mpi process addressing in mpi applications. In *Proceedings of the 29th European MPI Users' Group Meeting, EuroMPI/USA'22*, page 27–36, New York, NY, USA, 2022. Association for Computing Machinery. doi: 10.1145/3555819.3555829.
- Jean-Baptiste Besnard, Julien Adam, Allen D Malony, Sameer Shende, Julien Jaeger, Patrick Carribault, and Marc Pérache. Exploring space-time trade-off in backtraces. In *Tools for High Performance Computing 2018/2019*, pages 151–168. Springer, 2021. doi: 10.1007/978-3-030-66057-4_8.
- Julien Adam, Maxime Kermarquer, Jean-Baptiste Besnard, Leonardo Bautista-Gomez, Marc Pérache, Patrick Carribault, Julien Jaeger, and Allen D. Malony and Malon Sameer Shende. Checkpoint/restart approaches for a thread-based MPI runtime. *Parallel Computing*, 85:204–219, 2019. doi: 10.1016/j.parco.2019.02.006.
- Jean-Baptiste Besnard, Julien Jaeger, Allen D. Malony, Sameer Shende, Hugo Taboada, Marc Pérache, and Patrick Carribault. Mixing ranks, tasks, progress and nonblocking collectives. In *Proceedings of the 26th European MPI Users' Group Meeting, EuroMPI '19*, New York, NY, USA, 2019. Association for Computing Machinery. doi: 10.1145/3343211.3343221.
- Jean-Baptiste Besnard, Allen D. Malony, Sameer Shende, Marc Pérache, Patrick Carribault, and Julien Jaeger. Unifying the analysis of performance event streams at the consumer interface level. In Christoph Niethammer, Michael M. Resch, Wolfgang E. Nagel, Holger Brunst, and Hartmut Mix, editors, *Tools for High Performance Computing 2017*, pages 57–71, Cham, 2019. Springer International Publishing. doi: 10.1007/978-3-030-11987-4_4.
- Julien Adam, Jean-Baptiste Besnard, Allen D. Malony, Sameer Shende, Marc Pérache, Patrick Carribault, and Julien Jaeger. Transparent high-speed network checkpoint/restart in MPI. In *Proceedings of the 25th European MPI Users' Group Meeting*, Barcelona, Spain, September 23-26, 2018, pages 12:1–12:11, 2018. doi: 10.1145/3236367.3236383.
- Jean-Baptiste Besnard, Allen D. Malony, Sameer Shende, Marc Pérache, Patrick Carribault, and Julien Jaeger. Towards a better expressiveness of the speedup metric in MPI context. In *46th International Conference on Parallel Processing Workshops, ICPP Workshops 2017*, Bristol, United Kingdom, August 14-17, 2017, pages 251–260, 2017. doi: 10.1109/ICPPW.2017.45.
- Antoine Capra, Patrick Carribault, Jean-Baptiste Besnard, Allen D. Malony, Marc Pérache, and Julien Jaeger. User co-scheduling for mpi+openmp applications using openmp semantics. In *Scaling OpenMP for Exascale Performance and Portability - 13th International Workshop on OpenMP, IWOMP 2017*, Stony Brook, NY, USA, September 20-22, 2017, *Proceedings*, pages 203–216, 2017. doi: 10.1007/978-3-319-65578-9_14.
- Jean-Baptiste Besnard, Julien Adam, Sameer Shende, Marc Pérache, Patrick Carribault, and Julien Jaeger. Introducing task-containers as an alternative to runtime-stacking. In *Proceedings of the 23rd European MPI Users' Group Meeting, EuroMPI 2016*, Edinburgh, United Kingdom, September 25-28, 2016, pages 51–63, 2016. doi: 10.1145/2966884.2966910.
- Jean-Baptiste Besnard, Allen D. Malony, Sameer Shende, Marc Pérache, and Julien Jaeger. Gleaming the cube: Online performance analysis and visualization using malp. In Andreas Knüpfer, Tobias Hilbrich, Christoph Niethammer, José Gracia, Wolfgang E. Nagel, and Michael M. Resch, editors, *Tools for High Performance Computing 2015*, pages 53–66, Cham, 2016. Springer International Publishing. doi: 10.1007/978-3-319-39589-0_5.
- Jean-Baptiste Besnard, Allen D. Malony, Sameer Shende, Marc Pérache, Patrick Carribault, and Julien Jaeger. An MPI halo-cell implementation for zero-copy abstraction. In *Proceedings of the 22nd European MPI Users' Group Meeting, EuroMPI 2015*, Bordeaux, France, September 21-23, 2015, pages 3:1–3:9, 2015. doi: 10.1145/2802658.2802669.
- Jean-Baptiste Besnard. Profiling and debugging by efficient tracing of hybrid multi-threaded HPC applications. *Theses, Université de Versailles-Saint Quentin en Yvelines, July 2014*. doi: https://theses.hal.science/tel-01133344/file/2014VERS0007.pdf.
- Jean-Baptiste Besnard, Marc Pérache, and William Jalby. Event streaming for online performance measurements reduction. In *42nd International Conference on Parallel Processing, ICPP 2013*, Lyon, France, October 1-4, 2013, pages 985–994, 2013. doi: 10.1109/ICPP.2013.117.

Program Committees

2026	Reviewer , Performance Measurement, Modeling, and Tools, Supercomputing 2026	Conference
2026	Reviewer , Programming, Compilers and Performance, Euro-Par 2026	Conference
2026	Reviewer , Parallelism Track, Compass 2026	COMPASS
2025	Reviewer , Poster Track, HIPC 2025	Conference
2025	Reviewer , System Software Track, HIPC 2025	Conference
2025	Reviewer , Performance Track, Supercomputing India 2025	Conference
2025	Reviewer , System Software Stack Track, ICPP 2025	Conference
2025	Reviewer , Programming, Compilers and Performance, Euro-Par 2025	Conference
2025	Reviewer , DynResHPC	Euro-Par Workshop
2024	Reviewer , DynResHPC	Euro-Par Workshop
2024	Reviewer , Programming, Compilers and Performance, Euro-Par 2024	Conference
2024	Reviewer , System Software and Cloud Computing, SC24	Conference
2024	Co-Chair , Tech Paper Track Performance, ICCP 2024	Conference
2023	Reviewer , HPC Bug Fest 2023	SC23 Workshop
2023	Reviewer , ProTools 2023	SC23 Workshop
2023	Reviewer , Future Generation Computer Systems 2023 Journal	Journal
2023	Poster-Chair , EuroMPI 2023	Conference
2023	Co-Chair , Performance Track, SC23	Conference
2023	Reviewer , HPCMALL 2023	ISC23 Workshop
2023	Reviewer , Architectures and Accelerators, EuroPar 2023	Conference
2023	Reviewer , Evaluation, & Analysis track program committee, ISC High Performance Conference	Conference
2022	Reviewer , International Journal of High Performance Computing Applications	Journal
2022	Reviewer , ProTools 2022	SC22 Workshop
2022	Reviewer , Performance Measurement, Modeling, and Tools, SC 2022	Conference
2022	Reviewer , Future Generation Computer Systems 2022	Journal
2022	Reviewer , HPCMALL 2022	Workshop
2022	Reviewer , Performance Modeling and Evaluation track, CCGrid 2022	Conference
2021	Reviewer , Future Generation Computer Systems 2021	Journal
2021	Reviewer , ICPP 21	Conference
2020	Reviewer , ProTools 2020	SC20 Workshop
2020	Reviewer , HPCMaspa 2020	Workshop
2019	Reviewer , CANOPIE HPC 19	SC19 Workshop
2019	Reviewer , ProTools 19	SC19 Workshop
2019	Reviewer , HPCMaspa 2019	Workshop
2019	Reviewer , Performance Modeling and Evaluation track, CCGrid 2019	Conference
2018	Reviewer , 7th Workshop on Extreme-Scale Programming Tools	SC18 Workshop
2018	Reviewer , Posters, EuroMPI 2018	Conference
2018	Reviewer , Performance Track, ICPP 2018	Conference
2018	Reviewer , Performance Modeling and Evaluation track, CCGrid 2018	Conference
2017	Reviewer , HPCMaspa 2017	Workshop
2017	Reviewer , 6th Workshop on Extreme-Scale Programming Tools	SC17 Workshop
2017	Reviewer , Future Generation Computer Systems 2017	Journal
2017	Reviewer , Posters, EuroMPI/USA 2017	Conference
2017	Reviewer , Performance Measurement, Modeling, and Tools, SC 2017	Conference
2016	Reviewer , Performance Measurement, SC 2016	Conference
2016	Reviewer , HPCMaspa 2016	Conference